



DEPARTMENT OF COMPUTER SCIENCE

TDT4259 - APPLIED DATA SCIENCE

GROUP 17

---

# Student Dropout Detection

---

*Authors:*

Mats Bakketeig  
Christoffer Hafskjold  
Daniel Høyland  
Lara Pinheiro Jacobsen  
Aksel Matashev  
Hallvard Jensen  
Ådne Mauno

7th November 2025

---

# Contents

|                                                        |            |
|--------------------------------------------------------|------------|
| <b>List of Figures</b>                                 | <b>iii</b> |
| <b>List of Tables</b>                                  | <b>iv</b>  |
| <b>1 Introduction and Problem Definition</b>           | <b>1</b>   |
| 1.1 Introduction . . . . .                             | 1          |
| 1.2 Problem Definition . . . . .                       | 1          |
| 1.3 Team, Roles, and Responsibilities . . . . .        | 2          |
| <b>2 Background</b>                                    | <b>3</b>   |
| 2.1 Objectives . . . . .                               | 3          |
| 2.2 Problem approach . . . . .                         | 4          |
| 2.3 Examples of Data-Driven Impact . . . . .           | 5          |
| 2.4 Data Strategy . . . . .                            | 5          |
| 2.5 CRISP-DM . . . . .                                 | 6          |
| <b>3 Method and Analysis</b>                           | <b>8</b>   |
| 3.1 Dataset Description . . . . .                      | 8          |
| 3.2 Data Preprocessing . . . . .                       | 8          |
| 3.3 Data Analysis Tools and Methods . . . . .          | 10         |
| 3.3.1 Machine Learning Model Tools . . . . .           | 12         |
| 3.3.2 Methods for Data Analysis . . . . .              | 13         |
| 3.4 Evaluation Metrics . . . . .                       | 14         |
| <b>4 Evaluation and Interpretation</b>                 | <b>15</b>  |
| 4.1 Setup and Scoring Function . . . . .               | 15         |
| 4.2 Evaluation Metrics . . . . .                       | 15         |
| 4.3 Results and Performance Summary . . . . .          | 15         |
| 4.3.1 Technical results . . . . .                      | 15         |
| 4.3.2 Business results . . . . .                       | 16         |
| 4.4 Interpretation and Business Relevance . . . . .    | 18         |
| 4.4.1 Interpretation . . . . .                         | 18         |
| 4.5 Diagnostics, Robustness, and Limitations . . . . . | 19         |
| 4.5.1 Robustness and Limitations . . . . .             | 19         |
| 4.5.2 Shortcomings . . . . .                           | 19         |
| 4.5.3 Effect on Interpretation . . . . .               | 19         |

---

|                                            |           |
|--------------------------------------------|-----------|
| 4.6 Demo . . . . .                         | 19        |
| <b>5 Deployment and recommendations</b>    | <b>20</b> |
| 5.1 Deliverables . . . . .                 | 20        |
| 5.2 Target audience . . . . .              | 20        |
| 5.3 Implementation plan . . . . .          | 21        |
| 5.4 Recommended actions . . . . .          | 21        |
| 5.4.1 Academic cases . . . . .             | 22        |
| 5.4.2 Financial cases . . . . .            | 22        |
| 5.4.3 Sociodemographic cases . . . . .     | 22        |
| 5.4.4 Ambiguous cases . . . . .            | 23        |
| 5.5 Limitations . . . . .                  | 23        |
| 5.6 Future research . . . . .              | 23        |
| <b>6 Monitoring and Maintenance</b>        | <b>25</b> |
| 6.1 Monitoring plans . . . . .             | 25        |
| 6.2 Key Performance Indicators . . . . .   | 25        |
| 6.3 Dashboard . . . . .                    | 26        |
| 6.4 Maintenance . . . . .                  | 26        |
| 6.4.1 Alarms . . . . .                     | 27        |
| 6.5 Lessons Learned and Feedback . . . . . | 27        |
| <b>References</b>                          | <b>28</b> |
| <b>Appendix</b>                            | <b>30</b> |

---

# List of Figures

|    |                                                                                                          |    |
|----|----------------------------------------------------------------------------------------------------------|----|
| 1  | CRISP-DM Process Diagram . . . . .                                                                       | 6  |
| 2  | Distribution between dropout and graduated students. . . . .                                             | 8  |
| 3  | Nationality distribution in the dataset. . . . .                                                         | 9  |
| 4  | Nationality distribution excluding “Portuguese”. . . . .                                                 | 9  |
| 5  | Portuguese vs. others — counts and international share. . . . .                                          | 10 |
| 6  | Correlation heatmap of numeric features with the target variable. . . . .                                | 11 |
| 7  | Feature–Target Correlation (sorted by strength). . . . .                                                 | 12 |
| 8  | ROC curve ( $AUC \approx 0.947$ ) . . . . .                                                              | 16 |
| 9  | Test results . . . . .                                                                                   | 16 |
| 10 | Top dropout-related features . . . . .                                                                   | 18 |
| 11 | Monitoring dashboard showing model predictions and institutional KPIs for student dropout risk . . . . . | 26 |
| 12 | Proof-of-concept web demo for dropout prediction. . . . .                                                | 31 |

---

# List of Tables

|   |                                                     |    |
|---|-----------------------------------------------------|----|
| 1 | Team members and primary responsibilities . . . . . | 2  |
| 2 | Project Business Objectives . . . . .               | 3  |
| 3 | Project Technical Objectives . . . . .              | 3  |
| 4 | Dropout Factors . . . . .                           | 4  |
| 5 | Classification report . . . . .                     | 16 |
| 6 | Implementation plan . . . . .                       | 21 |
| 7 | Dataset Feature Description . . . . .               | 30 |

---

## Glossary

**Correlation** A statistical measure that expresses the strength and direction of a linear relationship between two variables. . iii, 10–13

**F1-Score** The F1-Score is a measure of a machine learning model's classification performance that combines precision and recall into a single metric. 14–16

**Precision** Answers the question, "Of all the positive predictions made, how many were actually correct?". 14

**Recall** Answers the question, "Of all the actual positive cases, how many did the model correctly identify?". 14

---

# 1 Introduction and Problem Definition

## 1.1 Introduction

Academia and higher education plays a central role in today's society by providing students with knowledge, skills, and expertise within societal problems and their job market [18]. But the growing dropout rate within higher education is a concern that could have lasting consequences on both an individual and societal level.

Dropping out has a lot of direct consequences for the individual. It could limit their opportunities, career prospects, and employability. It could also lower their potential earnings and productivity. Another potential consequence could be causing psychological problems such as lowering their self-confidence and potentially causing a less happy life. [4]

But dropping out doesn't just impact the individual, it also impacts society and academic institutions in various ways. It impacts parental institutions by potentially lowering their government funding or damaging their reputation or ranking [2]. It can also negatively impact society by causing a shortage of skilled and well-educated labor within the workforce, economic losses, and loss of investments within higher education [15].

Students drop out of their studies for various reasons, some of the more common reasons are economic difficulties, social situation, psychological or mental difficulties, overloaded workload, or academic difficulties [16]. There are many potential reasons for dropping out and they often have an origin within the student's private life and can therefore be quite difficult to identify. But this project will try to use economic, social and academic identifiers to map out the students most likely to drop out of their studies by training a machine learning system to calculate a likelihood of a given student dropping out of their study program based on these identifiers.

One instance where the dropout rate is way too high is within higher education in Portugal, where according to the "La Caixa" Foundation, the dropout rate for undergraduate students was as high as 30% in 2018 [5]. The purpose of this project is to assist the educational and academic institutions within higher education in Portugal with lowering this dropout rate to a more acceptable level, by developing a tool and a machine learning model to help identify students' likelihood of dropping out across different academic institutions. This tool will provide an opportunity for the institutions to first identify and thereafter offer appropriate guidance or support for the individuals close to dropping out.

## 1.2 Problem Definition

The main issue that this project is going to handle is the *early detection of students likely to drop out*. Even though universities gather a lot of administrative and academic data, many institutions fail to analyze these factors in combination effectively and thus do not proceed with prediction. Consequently, this leads to late interventions when students are already disengaged or financially unable to continue their studies.

Our project addresses two challenges that are interconnected:

1. **Identifying the significant factors contributing to dropout**, through examining the correlation between demographic, academic, financial, and socio-economic factors and student outcomes.
2. **Predicting students risk of dropping out**. Based on these factors and the correlation between them, find the students with the highest probability of dropping out.

---

### 1.3 Team, Roles, and Responsibilities

The team is comprised of seven students who come from various academic backgrounds, all informatics related but with different specializations. This provides the team with a diverse range of knowledge, experiences and skills, which allows the team to innovate a proper solution to the student dropout issue.

The team was separated into two smaller groups, one with primary responsibility for the report, and the other with primary responsibility for training and evaluating the machine learning model. But the entire group went together to understand the project and dataset, and to take the big project decisions, such as certain parts of feature engineering, report structure, etc.

| Student                | Academic background                      | Project responsibility                                                                                                                                |
|------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aksel Matashev         | BSc Programming and MSc Informatics      | Performing data analysis and creating model, and writing in report, especially Method and Analysis (Chapter 3)                                        |
| Christoffer Hafskjold  | BSc Programming and MSc Informatics      | Research and report writing, especially for Introduction (Chapter 1)                                                                                  |
| Ådne Mauno             | BSc Programming and MSc Informatics      | Research and report writing, especially for Background (Chapter 2)                                                                                    |
| Daniel Høyland         | BSc Programming and MSc Informatics      | Performing data analysis and creating model, and writing in report, especially Method and Analysis (Chapter 3) and Evaluation and results (Chapter 4) |
| Hallvard Jensen        | BSc Informatics and MSc Informatics      | Creating implementation plan and identify recommended actions (Chapter 5)                                                                             |
| Lara Pinheiro Jacobsen | BSc Informatics and MSc Informatics      | Contributing to Monitoring and Maintenance (Chapter 6) and creating the project wireframes                                                            |
| Mats Bakketeig         | BSc Computer Science and MSc Informatics | Analyzing model results in terms of business value and propose monitoring plan (Chapter 4 & 6)                                                        |

Table 1: Team members and primary responsibilities

The table (Table 1) displays what the primary responsibilities are for all team members. But their work is not limited to this area, every team member shared a common responsibility to complete the project on time and to correct any misconceptions and incorrections/mistakes within the group, project, or report.



---

## 2 Background

The background section will present the identified objectives of the project and describe the approach used to address the issue of student dropouts. The data strategy that will be used for this project will be presented as well as put emphasis on the impact data-driven development will have in mitigating the student dropout issue. Lastly the background section will show how the project used the CRISP-DM framework to connect the project business goals to data-mining.

### 2.1 Objectives

The main issue being handled in the project is the early detection of students likely to drop out of their studies, see section 1.2. After the main issue was identified, the early stages of the project phase was spent to research the data in the dataset, and identify underlying patterns and trends contributing to student dropout. The research part of the project was especially important, since it provided the team with valuable insight into dropout behaviour among students and helped define objectives which would be used as guidance for the project going forward.

To reflect the main issue, combined with findings of the research phase, the overarching goal of the project is to *develop a tool that allow educational institutions to detect and support students at the risk of drop-out.*

This goal is fulfilled through the completion of a predefined set of objectives. Below is a table for the identified business objectives for this project, see table 2.

| Business Objectives                                                                     | Description                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Identify students more likely to drop out of their studies</i>                       | To prevent a high dropout rate at academic institutions they have to first identify and thereafter provide avid support for students at risk of dropping out.                                                                                                                         |
| <i>Identify the main underlying factors contributing to student dropout probability</i> | To decide best action plan to prevent the student from dropping out, the cause or factors contributing to their increased risk of dropping out has to be.                                                                                                                             |
| <i>Describe practical strategies for intervention upon student dropout risk</i>         | To ensure students stay at their studies, the institution has to have clear action and intervention strategies to help in various situations and problems. These strategies are meant to support the student in various ways, including social, psychological, academic or financial. |

Table 2: Project Business Objectives

Accompanying these business goals, a set of technical objectives which the project will deliver on is defined in table 3.

| Technical Objectives                                                                | Description                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <i>Develop a predictive model to identify students at risk of dropping out</i>      | Train and validate a machine learning model to accurately detect students at the risk of dropout.                                 |
| <i>Make the model measure dropout probability</i>                                   | In addition to detecting students at the risk of dropout, the model should measure the risk as a prediction probability.          |
| <i>Make the model identify major factors contributing to dropout probability</i>    | In addition to providing the dropout probability, the model should provide the major factors influencing the dropout probability. |
| <i>Design a dashboard that displays student information and dropout probability</i> | Visualize prediction results in a clear and interpretable way so institutional staff can monitor student dropout probabilities.   |

Table 3: Project Technical Objectives

Achieving these objectives would allow institutions to anticipate and address student dropout risks

---

more effectively, making it possible to intervene early, resulting in drastically reducing the drop out rate among students.

## 2.2 Problem approach

In order to help educational institutions prevent or at least minimize student dropouts to the absolute minimum, the project will have an approach based on comprehensive research on causes related to student dropouts and predictive modeling techniques.

Before proceeding with the methods which will be used to tackle the issue of student dropout, the project first seeks to explore the student dropout issue in-depth, by reviewing relevant research literature that explains causes and impacts of student dropouts.

### Relevant Literature and Justification

Student dropouts is a problem that cannot be easily resolved and hence needs a comprehensive understanding and approach. According to a student dropout article, recording dropouts in Europe, dropout are performed either voluntarily or non-voluntarily. Voluntary dropouts reflects students choosing to leave their studies (switching careers, transfer universities, relocate, etc.,). Meanwhile, non-voluntary dropouts reflects students dropping out due to external factors (academic failure, financial difficulties, forced dismissal, etc.,). [8]. The same article references Tinto’s “Student Integration Model” [17], when presenting the fact that a student’s social and academic integration with the university is a crucial factor related to student dropouts. Another important aspect brought up by the research article is that early dropout (e.g., first year of study) happens more often than later dropout [8].

According to the same research article [8] there are nine distinct factors that stand as root causes for student dropout.

| Dropout Factor                                       | Description                                                                                                                 |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <i>Study conditions at university</i>                | The quality of teaching, curriculum, workload or institutional resources that affect the academic space.                    |
| <i>Academic integration at university</i>            | The extent of which students become engaged with their studies and academic system.                                         |
| <i>Social integration at university</i>              | The extent of which students become engaged with their social space, peers, teachers and general social life at university. |
| <i>Personal efforts and motivations for studying</i> | The students own drive to study, their study habits, time management, and management of university demands.                 |
| <i>Information and admission requirements</i>        | The requirements set for the university entry criteria.                                                                     |
| <i>Prior academic achievement in school</i>          | The student’s previous academic performances and how well prepared they are for higher education.                           |
| <i>Personal characteristics of the student</i>       | The student’s individual attributes, such as their age, gender, health, family, responsibilities, mental state, etc.,       |
| <i>Sociodemographic background of the student</i>    | The student’s socio-economic status, parental education, migrant background and other regional aspects.                     |
| <i>External conditions</i>                           | Factors outside the student’s control.                                                                                      |

Table 4: Dropout Factors

As observed from the research paper, both the financial and academic factors are especially important in regards to student dropout. The burden of financial obligations such as tuition fees or general financial expenses for required curriculum material has a heavy burden on the student’s shoulders. This would naturally be a much bigger issue for low-income socio-economics groups in the society. The dataset used for this project contains both academic and the financial factors, and is hence ideal for the development of models related to predicting student dropout probabilities.

---

Looking at the problem from the methodological angle, decision trees as well as ensemble methods have been the most common ones in educational data mining mainly because of their property of being able to deal with both categorical and numerical data without sacrificing the interpretability of the results [12]. Such models make it possible for the institutions not only to know the dropout rates beforehand but also to be aware of the factors responsible for that which is of utmost importance for the planning of the right interventions. Establishing our methodology on the already published classic ones, we are able to claim both the scientific rigor and the practical applicability.

## **Approach To Handling Data**

Initially, the data that will be used for this project, will first go through exploratory data analysis (EDA). This process is directly linked to the research phase, mentioned in section 2.1. The EDA process facilitates understanding of variable distribution, spotting of possible biases and identifying of missing or inconsistent data. At the end of this phase, the most important factors that predict students dropping out will be revealed, i.e., the tuition payment status, the first semester academic performance and the availability of scholarships.

After the dataset has been preprocessed, supervised machine learning methods will be applied to create predictive models. These models will be trained to label students as either "Dropout" or "Graduate" solely based on their attributes. Among the various models, decision trees, random forests, and boosting methods will be the front-runners because they can efficiently work on heterogeneous datasets full of categorical and numerical variables. The performance of the models will be checked through metrics like accuracy, precision, recall, and AUC with a strong focus on decreasing false negatives (students wrongly predicted to graduate), as these are the areas where interventions can come too late.

## **2.3 Examples of Data-Driven Impact**

Data-driven approaches has had a major impact on education by changing the way the educational institutions run their operations and even by impacting the decision of students to stay in school or drop out. One such case is the US Georgia State University, which turned to predictive analytics to keep tabs on the risk factors affecting 800 students [13]. This action that was taken, helped in the monitoring of students that were at risk of dropping out, putting in effort motivate students to not drop out of their studies. The program was effective in boosting graduation rates, especially among students with limited resources that have previous high dropout rates.

The open university in the UK used learning analytics to monitor student engagement with digital resources, and monitor how effective personalized tutoring were in mitigating student dropout in remote learning program [1].

The two mentioned cases shows proof that data-driven decision making in education can benefit students greatly, and outlines the potential this project can have on mitigating student dropouts. This project is intended to take a similar path as the two mentioned examples. The project will utilize a data strategy framework in combination with a supervised machine learning model to iteratively create a solution that makes it possible for the project to accomplish its objectives and predict students at risk of dropping out of their studies.

## **2.4 Data Strategy**

A data-driven approach will be leveraged to complete the objectives of the project. This means that a well designed data strategy that creates a systematic plan with the identified issue in focus throughout the development process of the selected solution, is important to have established.

The nature of the task for this project has many steps. Research must be done to understand business goals of educational institutions and data relevant to student dropouts predictability must be prepared and processed. A supervised machine learning model trained to predict student

---

dropout with the ability to clarify predictions in a understandable manner to educators must be made and constant evaluation of the models performance before deploying it as a service, means that a structured and iterative workflow is essential.

One of the well-known data-strategy frameworks is CRISP-DM (Cross-Industry Standard Process for Data Mining) and is the selected data-strategy framework for this project. Why CRISP-DM was selected and how it was utilized for this project is explained in section 2.5.

## 2.5 CRISP-DM

CRISP-DM is a data-driven framework that divides the development process into six phases used to help teams systematically work with data-related problems, while keeping the business goals at the very center throughout the entire project phase. Its one of the most common methodologies for data mining, analytics, and data science projects [19]. There is provided a CRISP-DM model that shows the sequential phases, see model 1.

There are a number of reasons why CRISP-DM was evaluated as the best data strategy framework for this project.

- Puts heavy emphasis on ensuring that the project aligns with the identified business goals and puts max effort on providing beneficial results for the organization which the project is made for.
- Has an iterative nature, that focuses on continuous evaluation and reflection of the results throughout the entire development process. In other words, puts emphasis on retrospective analysis to verify results against project objectives.

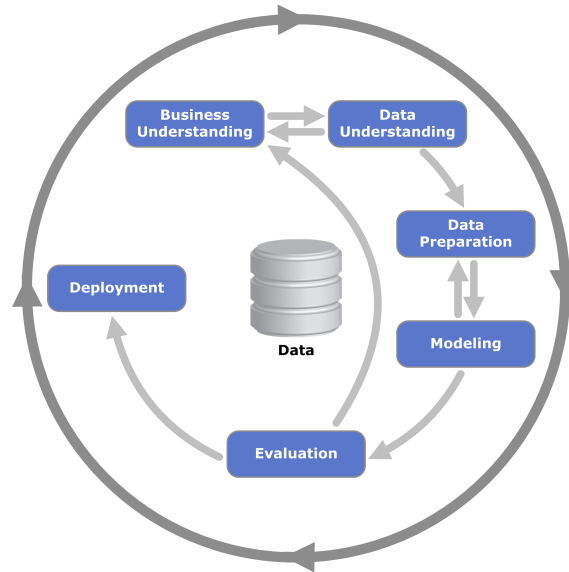


Figure 1: CRISP-DM Process Diagram

Source: Cross-industry standard process for data mining - Wikipedia [20]

Below are an explanation of the *CRISP-DM* phases and how they were used to organize the workflow of the project. For an in-depth explanation of the CRISP-DM phases, see the CRISP-DM article made by *Data Science PM* [19].

### Business Understanding

The business understanding phase is used to define the project's objectives and requirements from a business perspective. For this project, educational institutions were treated as the stakeholder

---

for the project with the business goal of mitigating students dropping out of their studies. Most of time during this phase was spent on researching reasons for why students drop out and draw similarities between the finding made by the team.

### **Data Understanding**

After the business phase, the data phase covers the process of data-gathering, researching and getting familiar with the data. The project concentrated on properly understanding the dataset with data reflecting factors related to student demographic information, academic performance indicators and enrollment details etc., and see what factors were crucial for what would estimate whether a student is in risk of dropping out or not.

### **Data Preparation**

This phase is used to process the data, creating a dataset suitable for training a model. Since the dataset was already cleaned of anomalies and null values by the dataset provider the team instead spent time seeing if it was necessary to clean more data or not. Other data processing techniques that were used were feature extraction and training the model where 80% of the dataset was used to train and 20% was set to the side for later performance testing.

### **Modeling**

The modeling phase covers the process where machine learning techniques and models are applied to the dataset. This phase started out with the team discussing which modeling techniques was better suited for this project. After coming to a conclusion on which modeling technique to use. Since the project uses CRISP-DM the modeling phase was an iterative process. Some dataset features were discussed whether they needed to be involved as well as to not include data not important for training the model

### **Evaluation**

After creating a model its important to evaluate its performance to ensure it meets both technical and business objectives. The evaluation of the model was an iterative process where the trained model trained on the remaining 20% of the dataset not used for training. The team decided to use common evaluation metrics as well as monitoring the model's performance using graphs. The results were compared to the business goals and discussed internally within the team to see if the results seemed to align well with what was expected.

### **Deployment**

The deployment phases defined the time and space where the implemented machine learning model is deployed in a real-world environment. The team decided pretty early on that there should be a dashboard which could present student information and display dropout probability with emphasis on user-friendliness so educators could understand the data. A trained machine learning model was integrated with a simple proof-of-concept, front-end application used internally within the team for testing purposes and visualizing the data. The team spent time discussing how the model would be used by the project stakeholder where the team landed on a solution to publish the model as a REST API, see section 5.1 for more information.

---

### 3 Method and Analysis

The dataset used in this study was collected from a higher education institution, containing students enrolled in various undergraduate degrees such as agronomy, design, education, nursing, journalism, management, social service, and technologies [14]. It contains information about each student, including demographic, academic, and socio-economic features, as well as students' academic performance at the end of the first and second semesters.

The main objective is to build classification models to predict whether a student will drop out or graduate, formulated as a binary classification problem. The dataset exhibits a strong class imbalance, with significantly more graduates than dropouts [14].

#### 3.1 Dataset Description

The dataset contains 4,424 instances with 36 features. The features are related to information about a student, their background and their academic performance. These features are described in Table 7.

Feature value descriptions were obtained from the original dataset documentation [14]. The target variable contains three labels: 'Graduate,' 'Dropout,' and 'Enrolled.' Only the first two were used in this analysis.

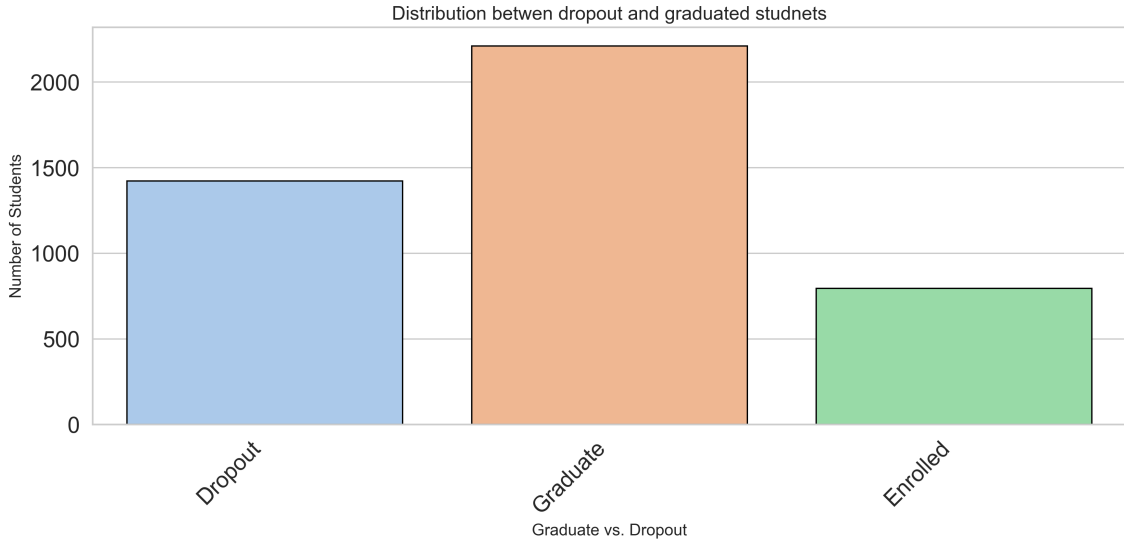


Figure 2: Distribution between dropout and graduated students.

As seen in Figure 2, the dataset has noticeably more graduates than dropouts, showing a class imbalance. This imbalance means the model could easily favor the majority class. It is also reflected in several features, such as gender and daytime/evening attendance. We try to minimize this imbalance and its bias in the model's predictions, as described in subsection 3.4.

#### 3.2 Data Preprocessing

The dataset has already been preprocessed by the authors where they did rigorous data preprocessing to handle data from anomalies, unexplainable outliers, and missing values. We further applied some processing.

First, we decided to drop the "Enrolled" data from the dataset since it wasn't something we wanted to predict. Including this class would bias the dataset and reduce prediction accuracy

for ‘Graduate’ and ‘Dropout’, since ‘Enrolled’ students will eventually move into one of those categories. So the data set went from 4,424 instances to 3,630 instances.

Since the dataset is relatively small, we performed a feature analysis and found that over 90% of the students were Portuguese as seen in the graph under Figure 3. The figures below exclude the ‘Enrolled’ category.

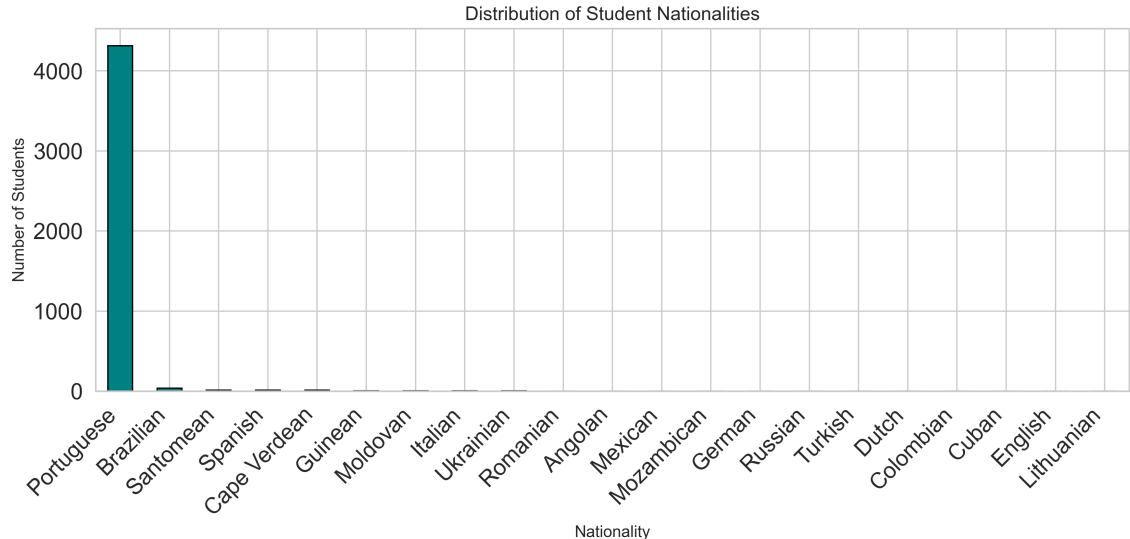


Figure 3: Nationality distribution in the dataset.

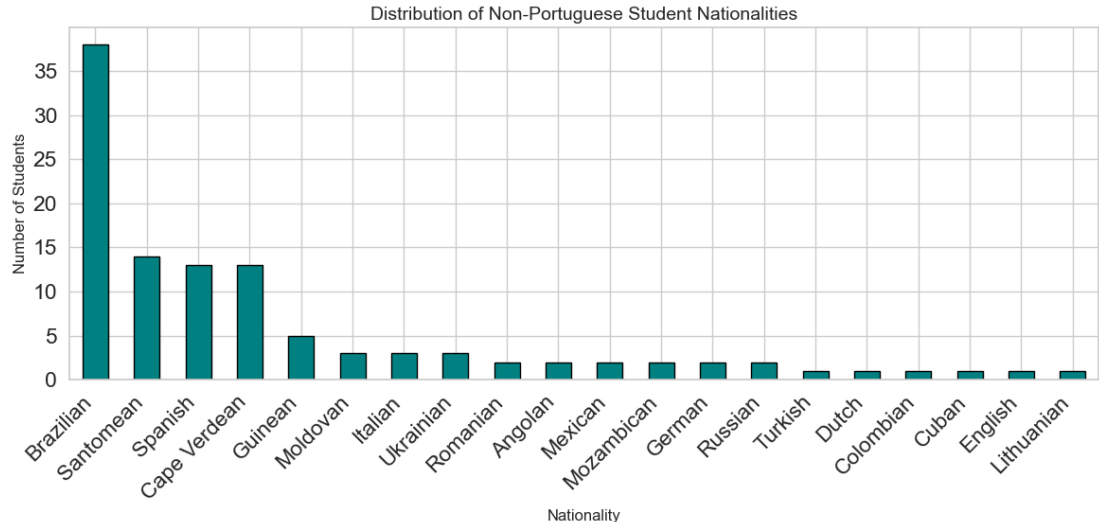


Figure 4: Nationality distribution excluding “Portuguese”.

Due to this large difference and the small number of students of other nationalities, we investigated whether the dataset originated from Portugal or if it simply contained a disproportionate number of Portuguese students.

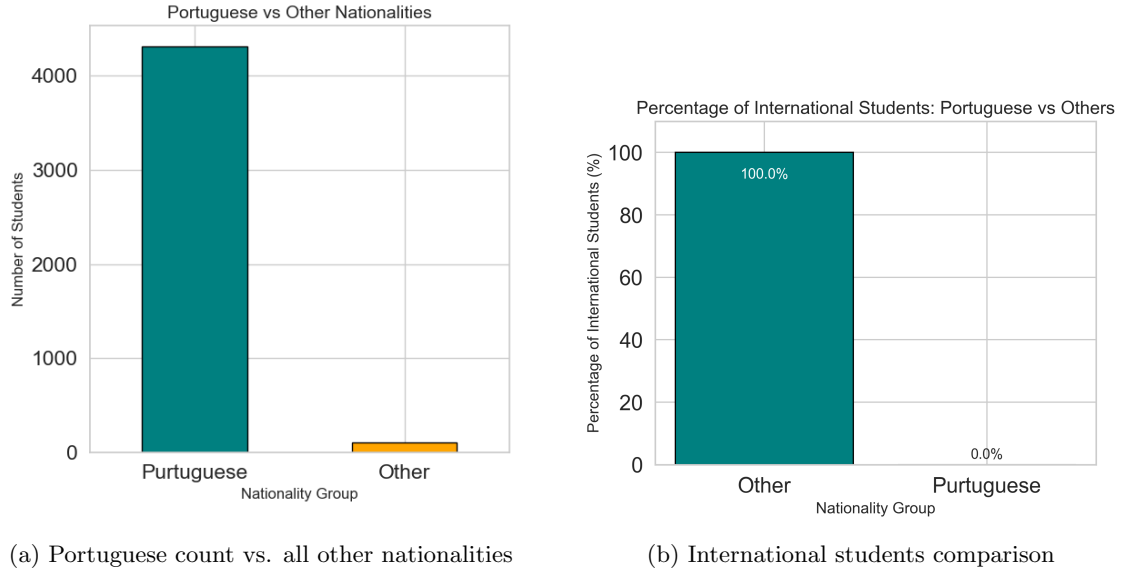


Figure 5: Portuguese vs. others — counts and international share.

As seen in Figure 5a, the difference is substantial. We compared the "other nationality" feature with the "International" and also did that to "Portuguese". As shown in Figure 5b, all Portuguese students were classified as non-international, while all other nationalities were international. This confirmed that the dataset represents a Portuguese institution. Since there were very few international students, all non-Portuguese nationalities were grouped under a single category labeled 'Other'.

### 3.3 Data Analysis Tools and Methods

The Correlation heatmap Figure 6 shows that the dataset contains several meaningful relationships between variables. These relationships indicate that the data holds usable signal for predicting student dropout. In the plot, red shades indicate positive and blue shades negative Correlations. Red features increase dropout likelihood, while blue features reduce it. The heatmap reveals several strong Correlations between academic performance variables, showing that the dataset contains meaningful signal for predicting student dropout.

A few features demonstrate the strongest Correlations with the target variable, *Target*, which represents whether a student drops out or graduates. Notably, Curricular units 2nd semester (approved) and Curricular units 2nd semester (grade) show the highest negative Correlations with dropout, meaning that higher academic achievement in the second semester is associated with a lower dropout risk. Other features, such as Debtor and Age at enrollment, display weak positive Correlations, suggesting that financial difficulties and age may slightly increase the likelihood of dropping out. The second plot Figure 7 summarizes the Correlation of each numeric feature with the target, helping identify which variables most influence the model. Overall, the analysis confirms that academic performance features contribute the strongest predictive signal for student outcomes.



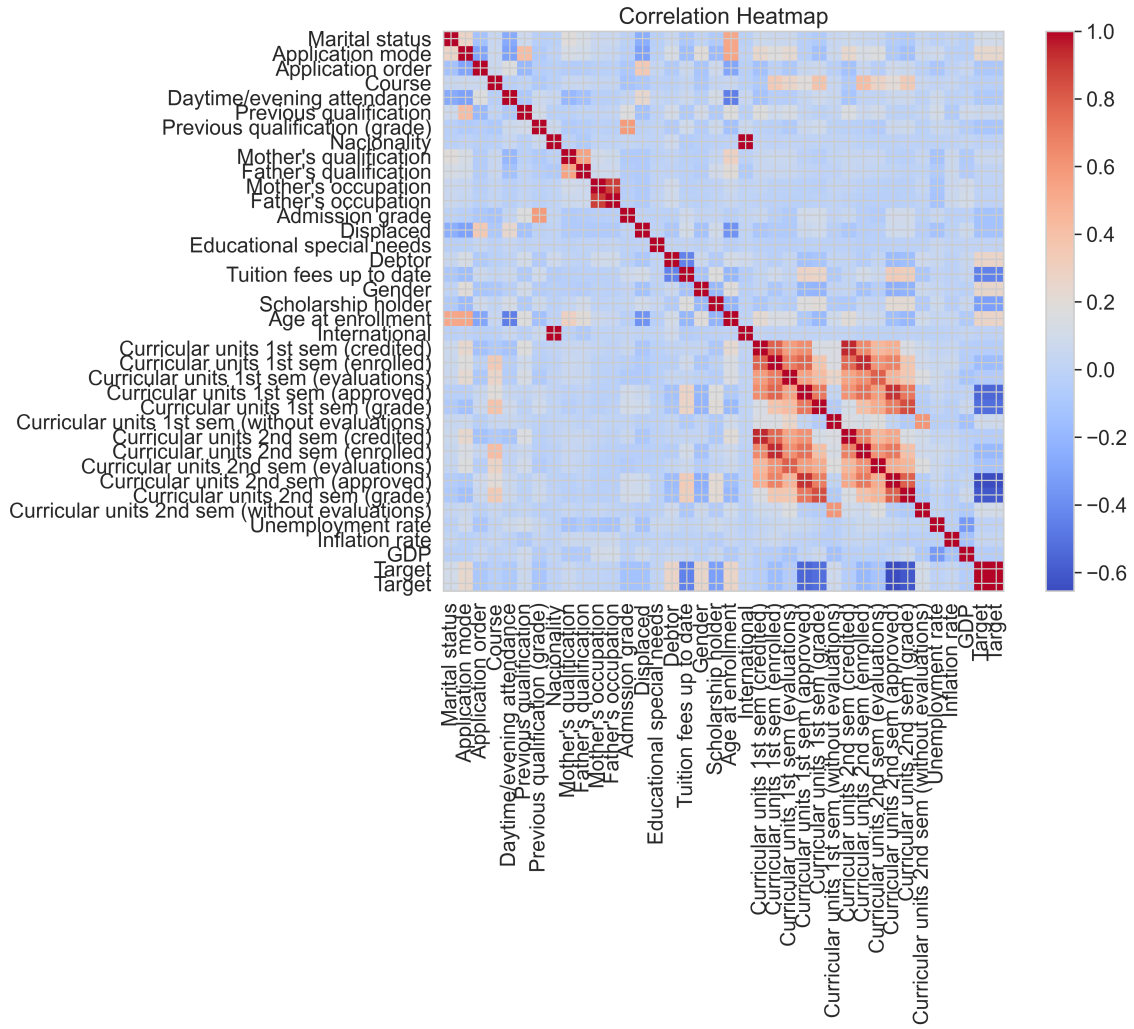


Figure 6: Correlation heatmap of numeric features with the target variable.

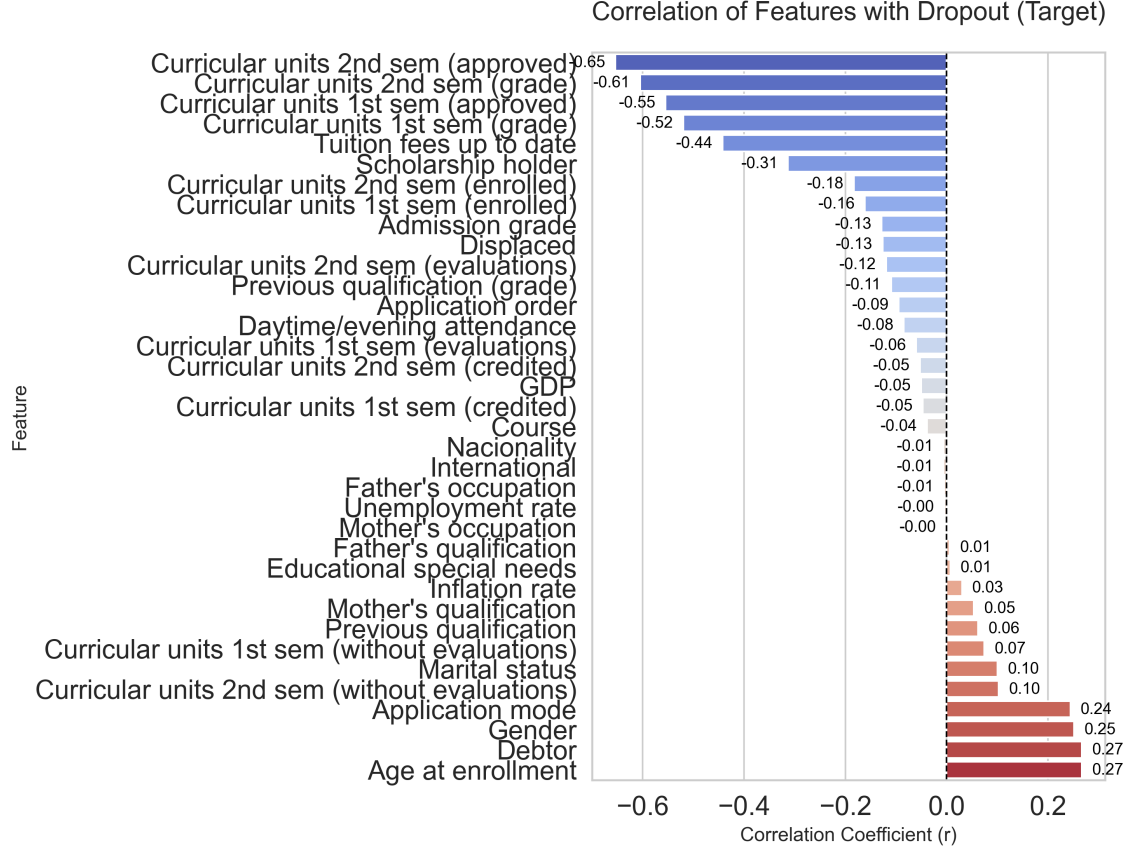


Figure 7: Feature–Target Correlation (sorted by strength).

$$r_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}$$

The above formula represents the *Pearson Correlation coefficient*  $r_{X,Y}$ , which measures the linear relationship between two variables  $X$  and  $Y$ . Values of  $r_{X,Y}$  close to 1 indicate a strong positive Correlation, values near  $-1$  a strong negative Correlation, and values around 0 suggest no linear relationship.

### 3.3.1 Machine Learning Model Tools

In this project, several Python-based tools and libraries were utilized for data processing, modeling, and evaluation.

#### Pandas

Was employed for data loading, cleaning, and manipulation. It provides high-level data structures like `DataFrame`, which simplifies operations such as filtering, grouping, and merging datasets.

#### NumPy

Supported efficient numerical computations and array operations, forming the foundation for many scientific and machine learning tasks. Arrays in NumPy allow element-wise operations and broadcasting, such as:

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \mathbf{b}$$

where  $\mathbf{X}$  is the feature matrix,  $\mathbf{w}$  the weight vector, and  $\mathbf{b}$  the bias term.

---

### Scikit-learn

Provided tools for preprocessing, data splitting, cross-validation, and evaluation metrics, ensuring consistent and reproducible model training workflows.

### Matplotlib

Used to create static visualizations such as histograms, scatter plots, and performance curves, while **Seaborn** enhanced these plots with statistical insights through heatmaps, boxplots, and Correlation matrices.

### ColumnTransformer

For preprocessing from the **scikit-learn** library, enables the combination of numeric and categorical transformation pipelines.

### StandardScaler

Used for numeric feature standardization, which transforms a feature ( $x$ ) into:

$$x' = \frac{x - \mu}{\sigma}$$

where  $\mu$  is the mean and  $\sigma$  is the standard deviation of the feature.

### OneHotEncoder

Categorical features were encoded into binary vectors using this, converting categories into numerical format suitable for model training.

### Logistic Regression

A manual implementation was developed to model binary outcomes. The hypothesis function used the sigmoid activation:

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

and parameters were optimized through gradient descent:

$$\theta := \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$$

where  $\alpha$  is the learning rate and  $J(\theta)$  is the cost function.

### StratifiedKFold Cross-validation

Was used to evaluate model performance. This ensured that each of the ( $k = 5$ ) folds preserved the original class distribution, providing a fair and balanced assessment of the classifier.

### 3.3.2 Methods for Data Analysis

Different machine learning methods were used to predict student outcomes as a part of the analysis:

1. **Logistic Regression:** Output probabilities for each class since it is logistic regression suitable for binary-class classification.
2. **Feature Scaling & Encoding:**

---

### 3.4 Evaluation Metrics

In order to determine the model performance in a very rigorous manner, the following strategies were put into practice:

- **Train-Test Split:** Stratified 80/20 division in order to maintain the same class distributions in both the training and test sets.
- **Cross-Validation:** 5-fold stratified cross-validation was performed to evaluate the model's stability and generalization across different subsets of the data.
- **Accuracy:** It calculates the ratio of correctly predicted student outcomes to all classes.
- **Precision, Recall, F1-Score:** Matrices handled their calculations class-wise to eliminate the effect of imbalances and to evaluate performance on minority classes.
- **Confusion Matrix:** It served the purpose of showing how model predictions compared to the actual outcomes.
- **ROC-AUC (One-vs-Rest):** It measures the model's capacity to sort classes based on the predicted probabilities.
- **Cross-Entropy Loss:** Used as the optimization objective during training to measure the difference between the predicted probabilities and the true class labels. Lower values indicate better model calibration and convergence toward accurate predictions.

This approach guarantees that the dataset, the preprocessing, the modeling, and the evaluation steps are not only well-defined but also reproducible, thus giving a solid base for the following analysis and interpretation of results.

---

## 4 Evaluation and Interpretation

This section evaluates the manually implemented logistic regression model used to predict whether a student will *drop out* or *graduate*.

### 4.1 Setup and Scoring Function

After preprocessing as explained in section 3.2, the dataset was divided into a training/test split. This resulted in two datasets where 80 % of the original dataset was allocated for training, while the remaining 20 % were used for testing. The reason for this was to test the model on previously unseen data, to simulate how the model would perform in a real-world scenario.

The model was trained using the training dataset for 5 000 epochs with a learning rate of 0.1. The model was then tasked to predict the target feature (whether or not the student would drop out) based on the input features provided in the test dataset. The prediction was then compared to the actual target value to determine whether or not the model made a successful prediction. All test results presented in this section is based on this comparison.

The prediction is based on the logistic (sigmoid) function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad P(\text{Dropout}) = \sigma(W_1X_1 + W_2X_2 + \dots + W_nX_n + b),$$

where  $P(\text{Dropout}) \in [0, 1]$  represents the predicted risk score.

### 4.2 Evaluation Metrics

Because the dataset is imbalanced, accuracy alone is insufficient to measure model performance. To account for this imbalance, other evaluation metrics have to be carefully chosen to effectively assess the model. Precision, recall, and F1-Score was thus selected to evaluate the model performance, and these are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Precision reflects the proportion of correctly identified dropouts among all predicted dropouts, while recall measures how many true dropouts were correctly detected. F1-Score on the other hand balances both.

### 4.3 Results and Performance Summary

#### 4.3.1 Technical results

The model achieved a test accuracy of 91 %, with detailed class-wise performance shown in table 5. Overall, the model demonstrates strong predictive ability: high recall for the *Dropout* class indicates that most at-risk students are correctly identified, while high precision suggests few false dropout alerts. The area under the ROC curve Figure 8 further confirms a strong ability to rank students by dropout risk. For instance, with a recall of 85 %, the model correctly detects approximately 85 out of 100 true dropouts.

---

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Graduate     | 0.91      | 0.95   | 0.93     | 442     |
| Dropout      | 0.91      | 0.85   | 0.88     | 284     |
| Accuracy     |           |        | 0.91     | 726     |
| Macro Avg    | 0.91      | 0.90   | 0.90     | 726     |
| Weighted Avg | 0.91      | 0.91   | 0.91     | 726     |

---

Table 5: Classification report

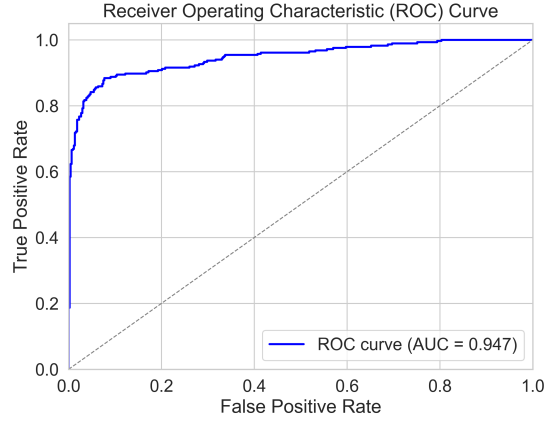


Figure 8: ROC curve (AUC  $\approx$  0.947)

#### 4.3.2 Business results

As discussed in section 4.3.1, the developed model is able to statistically determine whether a student is likely to drop out with a 91 % certainty. This value derived from the test results gathered when the model was tasked to make predictions on the test dataset (as explained in section 4.1). The test results are shown in figure 9.

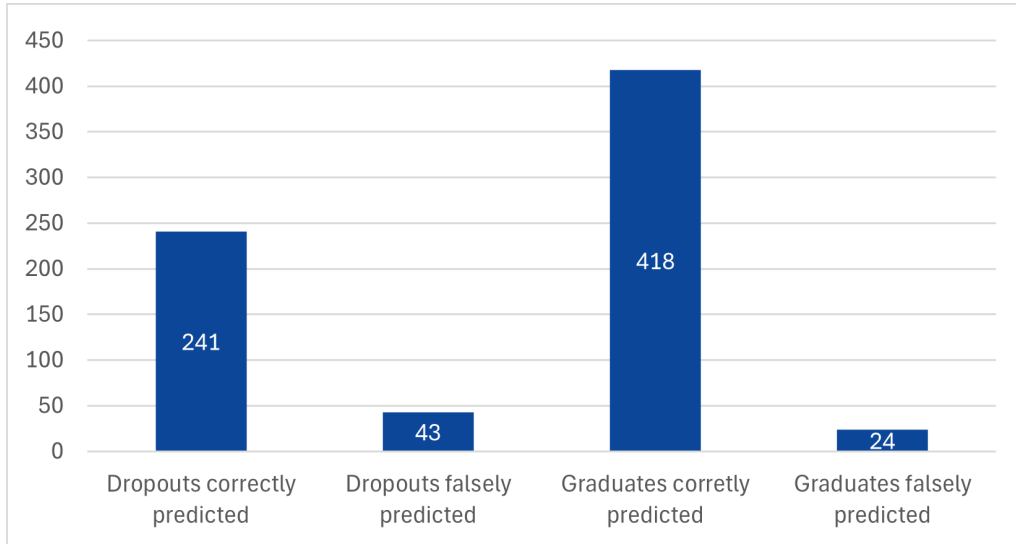


Figure 9: Test results

During testing the model attempted to predict whether each student in the test dataset was likely to drop out. Each test result thus represents the prediction on one student. The results were sorted into one of four categories:

- 
- **Dropouts correctly predicted:** Students likely to dropout that are predicted as potential dropouts by the model.
  - **Dropouts falsely predicted:** Students likely to dropout that are predicted as potential graduates.
  - **Graduates correctly predicted:** Students likely to graduate that are predicted as potential graduates.
  - **Graduates falsely predicted:** Student likely to graduate that are predicted as potential dropouts.

The test shows promising results, with the majority of students correctly predicted as a potential dropout or graduate. In total, the model made predictions on 726 students whereas 659 were classified correctly. Conclusive the model was able to correctly predict  $\approx 91\%$  of the students (explained in section 4.3.1).

### Relevant features

During testing the model was also able to determine which features contributed most to the test results. This means the model was able to tell what data gave a student a higher dropout risk after analyzing all students. Some of the top features include:

- Course
- Debtor
- Curricular units 2nd sem (enrolled)
- Curricular units 2nd sem (evaluations)

This shows that the biggest factor affecting whether a student is at risk of dropout is what course program they follow. Taking a diagnostic perspective, this suggests that student dropout is likely caused by inner factors related to each course program. Such factors may include the difficulty or structure of the course program. Following the course program factor is whether the student is in debt or not.

The next two features, as well as several other top features, are related to the curricular units of the student. The listed features include what units they are enrolled during the second semester as well as the amount of evaluations they have that same semester. This all indicates that academic factors generally have a bigger impact than demographic, financial, and socio-economic factors. The top features contributing to dropout risk are summarized in figure 10.

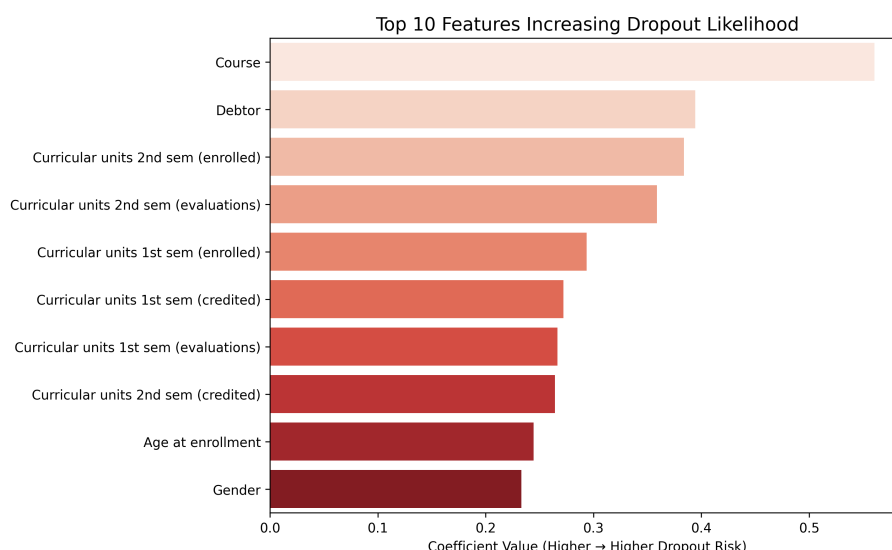


Figure 10: Top dropout-related features

Each feature is assigned a coefficient, measuring to what degree the features contribute to dropout risk (i.e. higher coefficient means higher contribution).

## 4.4 Interpretation and Business Relevance

Perhaps the most important part of any data analysis is the work done to bring real-world value from data. Value in the context of this project may be measured in terms of **business value**; how well the data contributes to the business objectives. Business value is only achieved when the data is transformed into valuable information.

### 4.4.1 Interpretation

The transformation from data to information is a process of interpretation; the data must be interpreted in terms of the business objectives to provide information to the business audience. The results provided in section 4.3 show examples of data to be interpreted.

#### Business objectives

*Identify students more likely to drop out of their studies.* As described in section 4.3, the developed model is able to determine whether a student is likely to drop out. Realistically, all students will have a probability of dropout, however students that are more likely can be identified by for example setting a probability threshold. Such a threshold would need to be determined along with business stakeholders, and will allow institution staff to more effectively detect at-risk students.

*Identify the main underlying factors contributing to student dropout probability.* Figure 10 shows the top features contributing to student dropout risk (explained in section 4.3.2). These features may be used as a base for solving this business objective. However, as will be discussed later in section 4.5.2, further analysis may be required to fully satisfy this objective.

*Describe practical strategies for intervention upon student dropout risk.* The results provided in section 4.3.2 allow analysis of contributing factors which may guide the steps for intervention. This will be discussed in detail in section 5.4.



---

## 4.5 Diagnostics, Robustness, and Limitations

The model exhibits two primary error patterns: false negatives and false positives. False negatives occur when actual dropouts are missed, often due to weak or delayed risk indicators. False positives arise when students show mixed signals, such as debt or age factors—but still maintain satisfactory academic performance.

### 4.5.1 Robustness and Limitations

The model’s performance remains stable across random splits, verified via 5-fold cross-validation, indicating good generalization. However, fairness metrics are less reliable due to limited representation of certain subgroups.

Including the Enrolled class might capture earlier dropout indicators. The small number of international students also introduces statistical uncertainty. Additionally, all features are static snapshots that may evolve over time, and correlation does not imply causation results should inform supportive interventions rather than punitive measures.

### 4.5.2 Shortcomings

Even though academic factors are stated to be the biggest contributing factors affecting whether a student will drop out (see section 4.3.2), this can only be considered a surface level view of the dataset. The data would have to be further analyzed to uncover any correlations, and see if academic performance is a product of causation from demographic, financial, or socio-economic factors. This can for example be done by completely overlooking academic factors (as dropouts seemingly has a poor academic performance), and only focusing on the other factors. For instance, the top non-academic contributing factor is *Debtor* (see figure 10), which is a financial factor.

It is also worth mentioning that the top contributing features are an aggregation of contributing features in the test dataset. The test dataset is, as explained in 4.1, only a simulated scenario. Once the system is put into effect the input data may highly differ from the test dataset, meaning that the stated top contributing features may change.

### 4.5.3 Effect on Interpretation

Excluding the data stated in section 4.5.1 makes the model more specific to its task and makes it less prone to bias. By including the removed *Enrolled* class the model would have the opportunity to detect potential dropouts earlier, but would lower the certainty of the predictions (as a new class needs to be predicted). This creates a trade-off between early and accurate predictions. For this data analysis, accurate predictions were considered more favorable in terms of the first business objective. This is also why potential biases were avoided.

In terms of the second and third business objective, the shortcomings stated in section 4.5.2 may have a more uncertain effect. It is difficult to measure the validity of the test data used to produce the results. If the test data does not reflect real-world input data well, institutions may end up focusing on the wrong indicators for dropout. This means that the model poorly identifies the underlying factors which also renders the steps for intervention unsatisfactory.

## 4.6 Demo

Using the trained model, a proof-of-concept web-based application was developed that allows users to input student attributes and receive a predicted dropout probability, the model’s confidence score, and the top five features most influential in the prediction. The demo interface is shown in the Appendix (Figure 12).

---

## 5 Deployment and recommendations

This chapter outlines how the dropout prediction model will be implemented in practice and how the results can be transformed into concrete actions within higher education institutions. The main deliverables will be presented, along with an accompanying implementation plan. When the implementation is in place, the recommended actions based on the model's predictions will be discussed. Finally, the limitations of the current solution are presented which will lay the foundation for future research.

### 5.1 Deliverables

The proposed model has up until now been used with the proof-of-concept web application. When the model is implemented at universities, the different institutions may have different existing systems they might want to integrate with, which is unknown at this stage. The most flexible solution is to publish an REST API which lets any application in the institution's ecosystem access the model. However, the chances are that many of the institutions does not have any obvious systems that are feasible to integrate with. Therefore, it is necessary to deliver a web-based front-end for the API as well.

The Flask server that was used for the demo can be delivered as the REST API. In the current iteration of the web server, the API accepts a JSON object with fields corresponding to that of the data set, as described in Table 7. The API can be further developed and extended with more endpoints if necessary, but the basic functionality is already in place.

An extension to the web service will be added before implementation, namely notification functionality. The web service will have a configurable percentage range for when to notify subscribers by means of Websockets. We opt for a configurable range because the institutions will have different amounts of resources available to handle student cases, and may have different policies governing whether to focus on the most at-risk students or the students that are most likely to be salvaged.

When new student data is uploaded, all subscribers are notified. This enables multiple stakeholders to be informed simultaneously of at-risk students such that action can be taken swiftly. All in all, this lets us integrate with a wide variety of systems that exists in the different institutions. The web service approach is the preferred approach for well established systems that requires as little disruption as possible.

A simple web front-end will be delivered to handle the cases where the API alone will not suffice. This will consist of an interface for requesting dropout probability for individual student, using a form similar to the demo, but also support for uploading spreadsheets containing information about multiple students. The spreadsheet must conform to a format that is expected by the application, and the user can download a template from within the application.

### 5.2 Target audience

The prediction of student dropout may be of interest for a wide range of people in a higher education institution. However, we limit the audience for this first iteration to faculty staff that deals with students, such as student advisors and counselors, but also the faculty administration. Student counselors and advisors are in direct contact with the students and has the necessary competence to assess the different situations that the students are in. This will enable them to make informed decisions based on the model results and the recommended actions that goes with it. Faculty administration is key to be able to resolve bigger picture issues that may come to light by means of our model.

Educational staff, such as lecturers and course coordinators are also important. They are even closer to the students and have great insight in what may be underlying causes for students risking dropout. For individual cases, there might be limited actions that can be implemented to

---

reduce dropout risk, but course structure, educational methods and curriculum is certainly key in influencing the wider student body.

Any technical staff will be essential, although they may not be the intended end-user. Since we plan to integrate with existing systems, the coordination with IT operators and maintainers is essential for our success. It is essential that direct communication with this group is maintained throughout the implementation plan.

### 5.3 Implementation plan

Since the solution can be deployed for any higher education institution (but especially applicable to Portuguese universities), the implementation plan is a general framework, which must be adapted to each university. The application will be hosted on a private cloud infrastructure, such as Amazon AWS, but this must be adapted from case to case, since the institutions will often require new software to be hosted on their existing infrastructure, either due to cost or policy. The plan structure is laid out in Table 6.

| Timeline  | Activity                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1-4  | <b>Preparation:</b> In the preparatory phase, dialogue is established with the institution. The main things to find out is what infrastructure the institution has that it wants to integrate with our solution, who will be the users of the solution, and who will administer the application when it has been deployed.                                                                               |
| Week 5-8  | <b>Site testing:</b> Users in the institutions is given access to the application to become familiar with the workflow. Any technical personnel is instructed about the solution’s API and architecture to best help them integrate with existing infrastructure. During this time the users and technical staff are invited to inquire about any aspects about the application that is unclear to them. |
| Week 9-10 | <b>Deployment:</b> The application put into production. Users start applying the solution in their workflows and the institution’s systems starts using the API. Necessary technical staff is still on premise to help with any early-stage challenges.                                                                                                                                                  |
| Onward    | <b>Monitoring and maintenance:</b> The model is monitored for alarms and errors. Maintenance is done as it is needed and any events that occur are handled. This phase continues for as long as the application is deployed, and is discussed in detail in section 6.                                                                                                                                    |

Table 6: Implementation plan

### 5.4 Recommended actions

Prediction of the dropout probability of a student, together with the configurable percentage range for alerts discussed in subsection 5.1, creates a binary condition for whether action should be taken. The model also identifies the five features that most strongly contributed to the prediction, providing guidance on which measures may be most effective.

Whenever student data leads to a dropout likelihood that falls within the target range, this triggers a call for action. Each institution may be differently organized, but these recommendations are based on the target audience described in subsection 5.2. A faculty official will assess each case to ensure the correct personnel are mobilized for the case. They will use the list of underlying factors listed by the model as foundation for this decision.

---

#### 5.4.1 Academic cases

For cases predicted based on primarily academic factors, such as grades, curricular units taken, and so on, student counselors will be engaged to aid the student in question. The student is invited to a meeting regarding their performance, focusing on confirming the model’s prediction and obtaining an overview of the student’s situation. The key is to get in contact with the student as fast as possible. As outlined in Figure 10, second semester enrollment and evaluations are detrimental to student dropout and indicates that early intervention is crucial to student retention.

When information is gathered from the student, actions should be implemented immediately. Student counselors should create an action plan for the student and schedule new follow-up appointments to discuss the effects of the actions. Furthermore, the counselor will make arrangements for tutoring, study groups, or extra lectures, depending on which measures are most useful to the student.

Although the strongest predictor of student dropout is the enrolled undergraduate program of the student (Figure 10), no recommendations can be put forward on organizational conditions at this point. Since the current model is trained to predict on a individual basis, actions aimed at faculty structure and organization is out of scope. Future research on this topic is essential.

All actions should be taken in accordance with the institutional privacy policies. Privacy of personal data such as predictions is essential, and only the necessary university staff should be given access. The data should only be used for student support work, as to not be susceptible to discriminatory or punitive measures.

#### 5.4.2 Financial cases

Financial cases are more difficult to handle than academic. Presumably, most universities do not have ample funds, and therefore there are few feasible actions to implement for students in this category. As discussed, structural conditions are out of scope for this solution, and so is political action. Nevertheless, it is crucial to address these issues, since student debtors are one of the groups that has highest risk of dropout.

According to [11] Portuguese citizens have a relatively low tuition cost of €697 for bachelor’s or master’s degrees. This may also apply to international students within the EU or EEA, but for the rest of the world, the price may be up to €7.000 [10]. Thus, financial difficulties are likely skewed towards international students. Since financial aid from the institutions is ruled out, alternative actions must take place.

If the student has tuition debts, the university can propose a payment plan for the student, if the university’s budget allows for it. This may give enough relief for the student to be able to get their finances in order and ultimately increase student retention. Another action that can be implemented is to recommend the student for scholarships. By giving a head start in the application process, this might help the student be selected for the scholarship. If the university gives out any scholarships of their own, they can prioritize students with financial challenges.

All financial cases should prompt the faculty to inform the students about any relevant student organizations that could be of assistance. Cheap student housing, meals, and student discounts should be all useful for the students, and the university can compile all the information in brochures.

#### 5.4.3 Sociodemographic cases

For cases related to social or demographic factors, such as gender, nationality, age, marital status etc., particular caution should be exercised by the institutions. These features are not actionable and should not be targeted directly. Instead, these factors should be used as indicators for certain groups of students that may need a more inclusive or supportive environment.

---

The faculty should identify indirect challenges of the different student groups. International students may face language barriers, cultural differences, or social difficulties. Thus, the faculty should focus on extra-curricular activities to help them socialize and arrange for language courses if possible. Older students that may have family obligations or are studying while working might benefit from more flexibility. The institutions can facilitate online classes, alternative study plans, or organize for studies that can be incorporated in their current job, if it is found relevant.

Actions based on sociodemographic cases should focus on creating equitable learning conditions rather than addressing the underlying factors themselves. This is especially important in preventing stigmatization and discrimination.

#### **5.4.4 Ambiguous cases**

If the cause for dropout is ambiguous, the faculty invites the student to a consultation to uncover the underlying challenges. With this information, it should be possible to apply the appropriate actions as discussed before.

### **5.5 Limitations**

Although the proposed solution demonstrates strong predictive performance, several limitations must be acknowledged. There are multiple challenges related to the data set, but also the model itself and its applications.

The first set of limitations relates to the geographic specificity and quality of the data. Most notably, the data was collected solely from Portuguese higher education institutions, as discussed. This made the solution best applied to Portuguese universities and colleges, since the model would reflect conditions at these institutions best. Furthermore, the data set is lacking in several areas. The model would benefit from having more features such as attendance, student satisfaction surveys, course evaluations, and features that give some insight into student mental health.

The data set also suffers from class imbalance, having a substantially larger amount of graduates than dropouts. Although measures were taken to mitigate the evaluation impact using precision, recall, and F1-score, the model will be inherently biased towards predicting graduates. However, one should keep in mind that graduate imbalance may be representative of reality, since most students will be graduating.

The cost imbalance is the practical result of the model bias caused by the class imbalance. Since the logistic regression model might have a bias toward predicting graduates, it is prone to false negative errors (i.e. missed dropout prediction). A false negative has a much higher institutional cost than false positives (i.e. incorrect dropout prediction) and represents a critical failure of the early warning system.

The static nature of the current solution severely affects the long-term viability. Without any mechanisms for periodic model retraining, the model will inevitably experience concept drift. Over time the predictive accuracy will degrade as the relationships between the student data and dropout risk changes.

The current model is also limited to predictions on an individual level. Consequently, it cannot be used to recommend actions on an organizational or structural level. As discussed earlier, the feature "Course" was shown to be the most significant factor contributing to student dropout probability. This is symptomatic of systemic problems with these programs and cannot be solved on an individual level.

### **5.6 Future research**

To overcome the acknowledged limitations of the current solution, future research should focus on three areas: data enhancement, model refinement, and expanding the actionable scope.

---

## **Data enhancement**

Future work must address geographic limitations and feature deficiency. Data collection should be expanded to incorporate educational institutions from as many regions of the world as possible. This will make the model more generalized and widely applicable. In addition, the feature set must be expanded beyond administrative records. This involves getting statistics about student health and psychological condition, which could be collected from attendance data and surveys. This will greatly improve the model, giving detailed insight into student characteristics that are unknown to the model as of now.

## **Modeling refinement**

The second of focus is related to the logistic regression's limitations in capturing the complexities of student behavior. Non-linear classification algorithms such as Random forests or Gradient Boosting should be considered, since this could enhance predictive performance. Gradient boosting can be used to minimize prediction errors, specifically for the minority class, by weighting misclassified data higher. Random Forests allows for better interpretation of feature significance, which is key for stakeholder's insights into the predictive factors [7]. Additionally, the model should experiment with recalibration by prioritizing the Recall metric of dropout to address the severe consequences of false negatives.

Crucial to the long-time viability of the solution is establishing a framework for automatic and periodic retraining of the model. This is essential to counteract concept drift and maintain predictive performance as institutions evolve. Retraining is also imperative to adapt to the effects of the actions taken based on previous predictions.

## **Expanding actionable scope**

Finally, the actionable scope of the model should be expanded beyond the individual-level predictions of the current model. By expanding to systemic diagnosis, the model can be used to address problems such as curriculum flaws for which the prevalence of the "Course" feature was symptomatic. Furthermore, organizational problems within institutions can be addressed, as well as program structure and educational methods. This can be achieved by expanding the data set even further and aggregating risk data based on factors such as course and department. Future solutions could include organizational alarm systems or dashboards for institutional level issues.

---

## 6 Monitoring and Maintenance

### 6.1 Monitoring plans

Monitoring plays a crucial role in the operations phase, ensuring a minimal amount of runtime issues. Focusing on continuous observation of the machine learning (ML) system's behavior to detect unwanted behavior, and enduring the ML models reliability, performance, and educational impact. One of the biggest concerns is model performance degradation, which is usually caused by data drift, concept drift or label drift [9].

By continuously monitoring data quality, model performance, and application outcomes through automated validation and audits each semester, the stakeholders will have the ability to ensure completeness, consistency, and relevance. Also, this while ensuring actions can be taken early to prevent degradation of model performance caused by data-related issues. This semi-automated monitoring ensures both the continuous monitoring, while keeping a human-in-the-loop [6].

Depending on what issue was detected, alerts and reports will be directed to different stakeholders. The technical alerts, such as drift detection or pipeline errors, will be handled by the data team, while application-level notifications about students or courses flagged as high risk will be directed to the counselors and educators.

### 6.2 Key Performance Indicators

The key performance indicator (KPI) is a performance metric used to monitor the ML system. Once deployed, the model will be continuously monitored using both technical and business KPIs.

#### Technical KPIs

From a technical perspective, the KPIs may be similar to the metrics used for evaluating the model (see section 4.2):

- Precision
- Recall
- F1-score

Recall is especially important for the *Dropout* class, as false negatives (students who are actually at risk but not identified as such) are missed opportunities for intervention. Operationally, the system will follow the share of high-risk students who persist after intervention, the overall retention rate vs. prior terms, and the conversion rate from "critical alerts" to counselor follow-ups.

#### Business KPIs

From a business perspective, institutional KPIs will be monitored to assess the real-world impact of the model. These would include:

- Amount of at-risk students
- Amount of critical cases
- Average risk score

In addition, the overall retention rate compared to previous semesters could serve as a down-the-line KPI once the system has been operational for a while. Different thresholds can be set to differentiate at-risk students from critical cases. In addition, an average risk score can be used as an indicator on the overall dropout risk among students in the institution. By using predictive accuracy with educational outcomes, the model's performance can be evaluated in both technical and social dimensions, ensuring that it corresponds meaningfully to institutional objectives.

### 6.3 Dashboard

The dashboard, illustrated in Figure 11, serves as the central monitoring interface for all stakeholders. It presents an overview of total students, at-risk cases, and critical alerts, while allowing users to see individual student profiles. Each record includes features such as attendance, GPA, and accumulated credits, providing the necessary context for institution staff to interpret risk levels and take informed action. Administrators can monitor institutional trends, while data teams can track model health and recalibration needs.

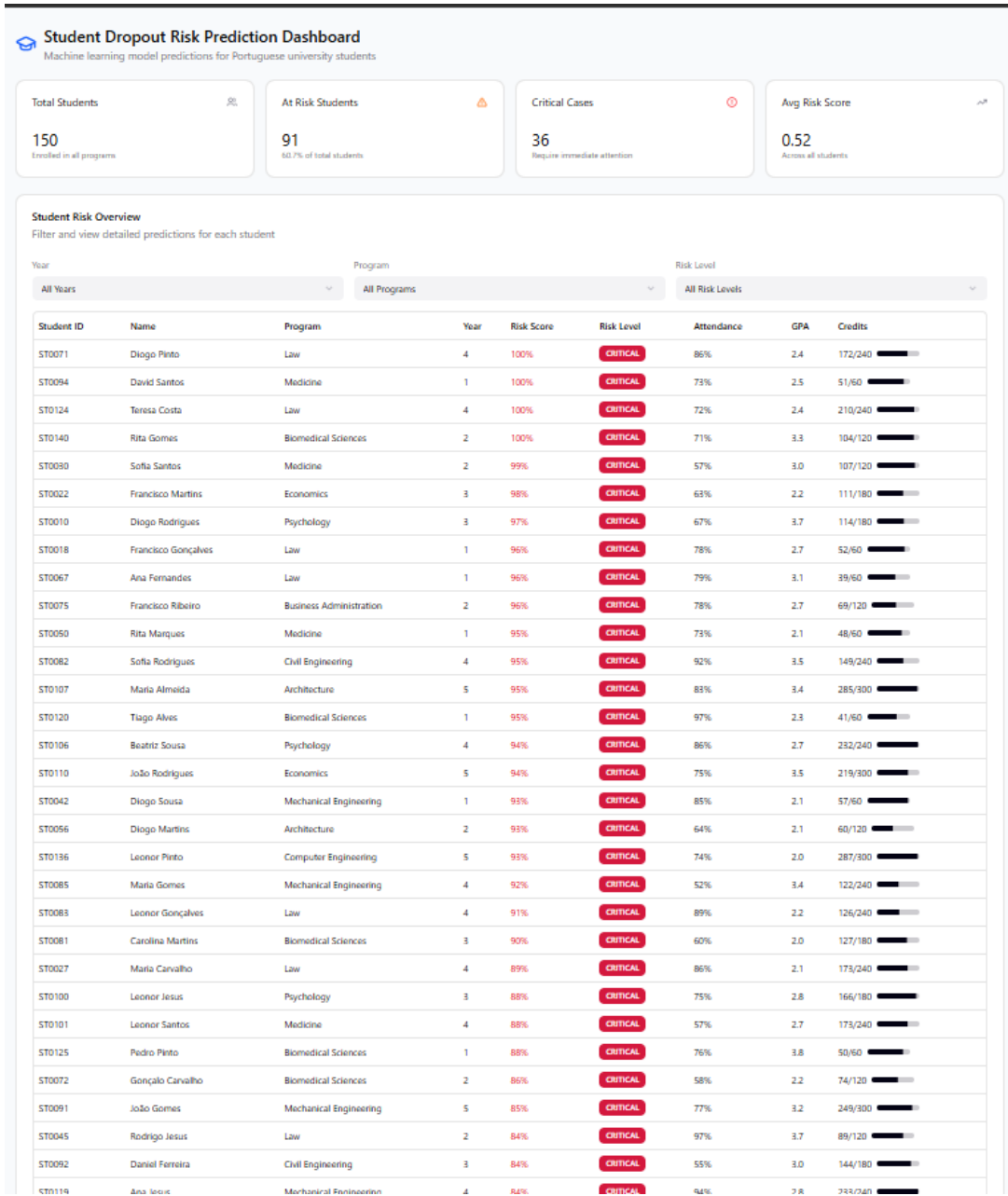


Figure 11: Monitoring dashboard showing model predictions and institutional KPIs for student dropout risk

### 6.4 Maintenance

The maintenance phase will focus on ensuring that the deployed system is sustainable, reliable, and maintains contextual relevance. Following the implementation plan mentioned in Section 5.3,



---

maintenance will include both technical upkeep of the model and continuous adaptation of the datasets due to the importance of up-to-date data [3].

Frameworks for monitoring data freshness will include automated alerts for data updates at the end of each semester, as well as establishing a comprehensive data life cycle management framework to allow tracking of the age and enforce policies for handling stale or obsolete data. This ensures that the model stays up to date and evolves alongside the changes in the institution, preserving relevance and accuracy.

#### 6.4.1 Alarms

Depending on the detected issue, alerts and reports will be directed to different stakeholders. Alarms and alerts have been put into three categories; **application-level**, **institutional-level**, and **system-level**, each serving a different purpose and notifying different stakeholders.

If a critical pattern in a student’s behavior is flagged as high risk, this will trigger an **application-level** alarm that will notify educators and administrators. This can include a sudden decline in a student’s average grade, repeated course failures, or change in attendance patterns suggesting disengagement. These alerts aim to prompt early institutional intervention before the student possibly drops out.

At the **institutional-level**, alarms may trigger when indicators surpass predefined thresholds, e.g., a sudden increase in predicted dropouts, or a specific course having an excessive amount of at-risk students. This helps institutions intervene both at the individual and systemic level, and supports proactive decision-making by the faculty and school administrators.

At the **system level**, model alarms will continuously monitor the technical health of the predictive model. Metrics such as precision, recall, and F1-score will be compared against baseline values established during initial deployment. A statistically significant decline in these metrics—typically a drop of 10 % or more from the baseline—will trigger an alert, prompting review and potential retraining.

As mentioned in Section 6.1, data quality and model performance will be monitored through automated validation. These alarms are used to detect performance degradation due to evolving data patterns or structural changes. When triggered, these alerts will automatically notify the data science team and log events in the model registry, ensuring full traceability and reproducibility of corrective actions.

### 6.5 Lessons Learned and Feedback

This data science project challenged the group to use data science in a way that goes beyond the application of statistical models. Even though several group members have various technical backgrounds more or less related to data science, everyone had to confront their previous knowledge and question if what they were doing was actually beneficial.

The implementation of the CRISP-DM framework has taught the group to work iteratively in a data science context, and how to bring business needs into the process. The key takeaway from this project has been this business-centered perspective, where all actions taken in the project should aim to create business value. Whether or not the group members will continue to pursue data science, the new perspectives gained from this project are invaluable for their role as developers.

---

## References

- [1] Avinash Boroowa and Christothea Herodotou. ‘Learning Analytics in Open and Distance Higher Education: The Case of the Open University UK’. In: 2022, pp. 47–62.
- [2] Edwin Buenaño, María José Beletanga and Mónica Mancheno. *What Factors are Relevant to Understanding Dropout? Analysis at a Co-Financed University in Ecuador and Policy Implications, Using Survival Cox Models*. 2023. URL: <https://www.tandfonline.com/doi/full/10.1080/15348431.2023.2271570#abstract>.
- [3] Abigail Collins and Antony Owen. ‘Data Quality Monitoring in MLOps’. In: (Nov. 2021).
- [4] Metropolitan State University of Denver. *10 Benefits of Higher Education*. 2022. URL: <https://www.msudenver.edu/10-benefits-of-higher-education/>.
- [5] ”La Caixa” Foundation. *Dropout in higher education: sociodemographic, economic, and psychosocial factors in the post-pandemic era*. 2024. URL: <https://observatoriosocial.fundacaolacaixa.pt/en/-/dropout-in-higher-education-sociodemographic-economic-and-psychosocial-factors-in-the-post-pandemic-era>.
- [6] Antonio A. Ginart, Martin Jinye Zhang and James Zou. ‘MLDemon: Deployment Monitoring for Machine Learning Systems’. In: *arXiv preprint arXiv:2208.12345* (2022).
- [7] Aravind Kumar Kalusivalingam, Neha Patel Amit Sharma and Vikram Singh. ‘Leveraging Random Forests and Gradient Boosting for Enhanced Predictive Analytics in Operational Efficiency’. In: *International Journal of AI and ML* 3.9 (2022). URL: <https://cognitivecomputingjournal.com/index.php/IJAIML-V1/article/view/72>.
- [8] Barbara Kehm, Malene Larsen and Hanna Sommersel. ‘Student dropout from universities in Europe: A review of empirical literature’. In: *Hungarian Educational Research Journal* 9.2 (2019). Accessed: 2025-11-05, pp. 147–164. URL: <https://akjournals.com/view/journals/063/9/2/article-p147.xml?body=fullHtml-28680>.
- [9] Grace A. Lewis et al. ‘Augur: A Step Towards Realistic Drift Detection in Production ML Systems’. In: *Proceedings of the IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*. Carnegie Mellon University Software Engineering Institute. Pittsburgh, PA, USA, 2021.
- [10] University of Lisboa. *Access and Admission 2022/23 International Student*. 2022. URL: <https://www.ulisboa.pt/sites/default/files/documents/page/2025/flyerinternationalstudent2223.pdf>.
- [11] University of Lisboa. *Tuition Fees*. 2025. URL: <https://www.ulisboa.pt/en/info/tuition-fees>.
- [12] Vasiliki Matzavela and Efthymios Alepis. ‘Decision tree learning through a Predictive Model for Student Academic Performance in Intelligent M-Learning environments’. In: *Computers Education* (Oct. 2021).
- [13] Michael Oppong. *Leveraging Predictive Analytics to Enhance Student Retention: A Case Study of Georgia State University*. 2024.
- [14] V. Realinho et al. *Predict Students’ Dropout and Academic Success [Dataset]*. UCI Machine Learning Repository. 2021. DOI: 10.24432/C5MC89. URL: <https://doi.org/10.24432/C5MC89>.
- [15] Vesna Skrbinjek, Dušan Lesjak and Valerij Dermol. *Higher Education Dropout: A Literature Review*. 2024. URL: <https://toknowpress.net/ISBN/978-961-6914-31-4/104.pdf>.
- [16] Indeed Editorial Team. *8 types of university dropout reasons and how to manage them*. 2025. URL: <https://uk.indeed.com/career-advice/career-development/university-dropout-reasons>.
- [17] Vincent Tinto. *Leaving college : rethinking the causes and cures of student attrition*. Chicago : University of Chicago Press, 1987.
- [18] UNESCO. *What you need to know about higher education*. 2025. URL: <https://www.unesco.org/en/higher-education/need-know>.
- [19] *What is CRISP DM?* [Online; accessed 4-November-2025]. 2024. URL: <https://www.datascience-pm.com/crisp-dm-2/>.
- [20] Wikipedia contributors. *Cross-industry standard process for data mining — Wikipedia, The Free Encyclopedia*. [Online; accessed 4-November-2025]. 2025. URL: [https://en.wikipedia.org/w/index.php?title=Cross-industry\\_standard\\_process\\_for\\_data\\_mining&oldid=1315954212](https://en.wikipedia.org/w/index.php?title=Cross-industry_standard_process_for_data_mining&oldid=1315954212).



## Appendix

| Feature                                              | Description                                                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Marital Status                                       | Student's marital status (1 – single, 2 – married, 3 – widower, 4 – divorced, 5 – facto union, 6 – legally separated).         |
| Application Mode                                     | Mode of application to the institution (e.g., general contingent, transfer, international student, special contingents, etc.). |
| Application Order                                    | Order of application preference (0 = first choice, up to 9 = last choice).                                                     |
| Course                                               | Code identifying the enrolled undergraduate program (e.g., Agronomy, Nursing, Management, etc.).                               |
| Daytime/Evening Attendance                           | Attendance type (1 – daytime, 0 – evening).                                                                                    |
| Previous Qualification (Grade)                       | Grade of previous qualification (scale: 0–200).                                                                                |
| Admission Grade                                      | Admission grade at enrollment (scale: 0–200).                                                                                  |
| Age at Enrollment                                    | Age of the student at the time of enrollment.                                                                                  |
| Gender                                               | Student's gender (1 – male, 0 – female).                                                                                       |
| Nationality                                          | Student's nationality (coded values representing countries).                                                                   |
| Displaced                                            | Whether the student lives away from home for study purposes (1 – yes, 0 – no).                                                 |
| Educational Special Needs                            | Indicates if the student has educational special needs (1 – yes, 0 – no).                                                      |
| Debtor                                               | Indicates if the student owes tuition or fees (1 – yes, 0 – no).                                                               |
| Tuition Status                                       | Whether tuition fees are up to date (1 – yes, 0 – no).                                                                         |
| Scholarship Holder                                   | Whether the student receives a scholarship (1 – yes, 0 – no).                                                                  |
| International Student                                | Whether the student is an international student (1 – yes, 0 – no).                                                             |
| Mother's Qualification                               | Mother's highest qualification level (coded educational levels).                                                               |
| Father's Qualification                               | Father's highest qualification level (coded educational levels).                                                               |
| Mother's Occupation                                  | Mother's occupation type (coded occupational categories).                                                                      |
| Father's Occupation                                  | Father's occupation type (coded occupational categories).                                                                      |
| Curricular Units – 1st Semester: Credited            | Number of curricular units credited in the 1st semester.                                                                       |
| Curricular Units – 1st Semester: Enrolled            | Number of curricular units enrolled in the 1st semester.                                                                       |
| Curricular Units – 1st Semester: Evaluations         | Number of curricular unit evaluations taken in the 1st semester.                                                               |
| Curricular Units – 1st Semester: Approved            | Number of curricular units successfully passed in the 1st semester.                                                            |
| Curricular Units – 1st Semester: Average Grade       | Average grade of curricular units in the 1st semester (scale: 0–20).                                                           |
| Curricular Units – 1st Semester: Without Evaluations | Number of curricular units without evaluations in the 1st semester.                                                            |
| Curricular Units – 2nd Semester: Credited            | Number of curricular units credited in the 2nd semester.                                                                       |
| Curricular Units – 2nd Semester: Enrolled            | Number of curricular units enrolled in the 2nd semester.                                                                       |
| Curricular Units – 2nd Semester: Evaluations         | Number of curricular unit evaluations taken in the 2nd semester.                                                               |
| Curricular Units – 2nd Semester: Approved            | Number of curricular units successfully passed in the 2nd semester.                                                            |
| Curricular Units – 2nd Semester: Average Grade       | Average grade of curricular units in the 2nd semester (scale: 0–20).                                                           |
| Curricular Units – 2nd Semester: Without Evaluations | Number of curricular units without evaluations in the 2nd semester.                                                            |
| Unemployment Rate (%)                                | National unemployment rate at the time of enrollment (percentage).                                                             |
| Inflation Rate (%)                                   | National inflation rate at the time of enrollment (percentage).                                                                |
| GDP                                                  | Gross Domestic Product value corresponding to the enrollment period.                                                           |
| Target                                               | Final student status (categorical: Dropout, Enrolled, or Graduate).                                                            |

Table 7: Dataset Feature Description

Student Graduation Prediction

Marital Status

Single

Application Mode

1st phase - general contingent

Application Order (0-9)

4

Course

Biofuel Production Technologies

Daytime/Evening Attendance

Daytime

Previous Qualification

Secondary Education

Previous Qualification Grade (0-200)

100

Admission Grade (0-200)

190

Age at Enrollment

21

Gender

Male

Displaced

Yes

Educational Special Needs

Yes

Debtor

Yes

Tuition Fees Up To Date

Yes

Scholarship Holder

Yes

Nationality

Portuguese

Mother's Qualification

Secondary Education - 12th Year

Father's Qualification

Secondary Education - 12th Year

Mother's Occupation

Student

Father's Occupation

Student

Curricular Units – 1st Semester

Credited

1

Enrolled

1

Evaluations

1

Approved

1

Average Grade (0-20)

18

Without Evaluations

0

Curricular Units – 2nd Semester

Credited

11

Enrolled

1

Evaluations

1

Approved

1

Average Grade (0-20)

19

Without Evaluations

0

Unemployment Rate (%)

21.7

Inflation Rate (%)

-0.7

GDP

1.1

Predict

Predicted Outcome: Graduate

Probability: 84.44%

Top 5 Dropout Risk Factors:

- 2Nd Semester (Approved)
- 2Nd Semester (Credited)
- 1St Semester (Approved)
- Debtor
- Gender

Figure 12: Proof-of-concept web demo for dropout prediction.